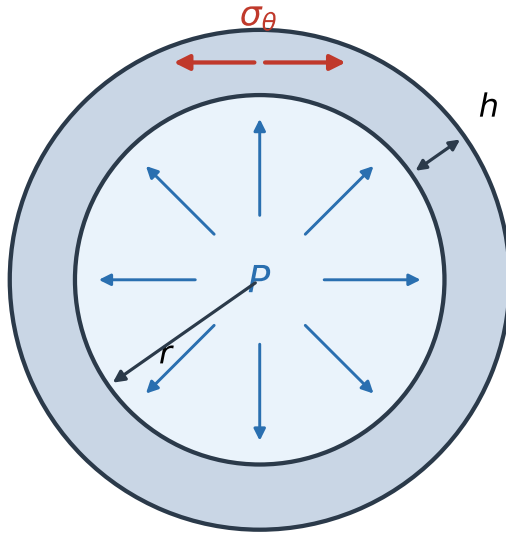


Shared setting: $R = 1.0 \text{ mm}$, $H = 130 \mu\text{m}$, $P_h = 13.3 \text{ kPa}$, $\bar{\sigma}_h = 102 \text{ kPa}$

(a) thin-walled artery (cross-section)

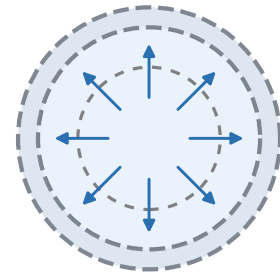
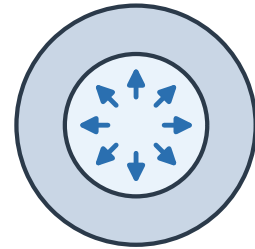


Laplace: $\sigma_\theta = Pr/h \propto \lambda^2$

$\odot z, \lambda_z$

(b) the two insults

hypertension: $P \uparrow \Rightarrow$ wall thickens



aneurysm: elastin lost $\Rightarrow$ dilation